

Ashutosh Pattanaik

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PROFESSIONAL SUMMARY

DevOps Engineer with hands-on experience in CI/CD automation, AWS cloud infrastructure provisioning, and container orchestration using Docker and Kubernetes. Skilled in Infrastructure as Code (Terraform, Ansible), Python scripting for automation, and integrating code quality tools into delivery pipelines. Passionate about automation and continuous improvement of software delivery workflows.

PROJECTS

CI/CD Pipeline – Java Web Application | *Jenkins, Docker, Kubernetes, SonarQube, Nexus, Tomcat, MySQL, Git, Python*

- Implemented a fully automated CI/CD pipeline with Jenkins, reducing manual deployment effort by ~80% and cutting average release cycle time from days to under 30 minutes.
- Integrated SonarQube quality gates, blocking 100% of non-compliant builds from reaching production and improving code quality scores by 35% over the internship period.
- Automated Docker image builds and registry pushes on every Git commit, achieving a fully touchless release process — eliminating ~10 manual steps per deployment.
- Deployed containerized applications to Kubernetes with rolling update strategy, achieving zero-downtime releases across 3+ production deployments and maintaining 99.9% uptime.
- Wrote Python scripts to automate post-deployment health checks and log parsing, reducing manual verification effort after each release.
- Diagnosed and resolved 15+ pod scheduling failures, image pull errors, and networking issues across a multi-node Kubernetes cluster, reducing mean time to recovery (MTTR) by 60%.

Two-Tier Containerized Application | *Docker, Docker Compose, MySQL, Nginx*

- Designed a two-tier containerized stack (web app + MySQL) with a custom Docker bridge network, reducing inter-container latency by isolating services and eliminating unnecessary network hops.
- Orchestrated multi-container deployment using Docker Compose, cutting environment setup time from ~45 minutes to under 2 minutes with a single command startup.
- Configured persistent Docker volumes for MySQL, ensuring 100% data durability across container restarts, and set up Nginx as a reverse proxy eliminating direct port exposure.
- Validated end-to-end connectivity between application and database layers with 0 data loss across 20+ container restart cycles during testing.

3-Tier AWS Architecture – Terraform & Ansible | *AWS (EC2, VPC, ALB, RDS, IAM), Terraform, Ansible, Python*

- Designed and provisioned a scalable 3-tier AWS architecture using Terraform modules, reducing infrastructure provisioning time by 70% compared to manual console setup.
- Configured an Application Load Balancer (ALB) distributing traffic across EC2 instances in 2 availability zones, achieving high availability with automatic failover in under 30 seconds.
- Automated configuration management across all EC2 tiers using Ansible playbooks, ensuring 100% environment consistency and eliminating configuration drift across 10+ servers.
- Used Python scripting for routine server tasks such as service status checks and log cleanup, complementing Ansible playbooks for day-to-day operations.
- Enforced AWS security best practices — least-privilege IAM roles, security group segmentation, and private subnet database isolation — reducing the attack surface by restricting 100% of direct DB internet access.

TECHNICAL SKILLS

Cloud: AWS – EC2, VPC, Subnets, Security Groups, ALB, Auto Scaling, RDS, IAM

CI/CD: Jenkins (pipeline-as-code, webhooks, multibranch pipelines)

Containers & Orchestration: Docker, Docker Compose, Kubernetes (Deployments, Services, ConfigMaps, Rolling Updates)

IaC & Config Management: Terraform, Ansible

Programming & Scripting: Python

Code Quality & Artifacts: SonarQube, Nexus Repository Manager

Servers, Build & VCS: Apache Tomcat, Nginx, Maven | Git, GitHub

Databases & OS: MySQL, Oracle | Linux (Ubuntu/CentOS), Windows

EDUCATION

B.Tech – Electronics & Telecommunication Engineering

2020 – 2024

Parala Maharaja Engineering College, Odisha

Diploma – Electronics & Telecommunication Engineering

2017 – 2020

UCPES, Odisha